

FOR TEP4175/FENT2321

Tips and rules on report writing

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Accreditation

Some sentences here and there have been taken from Simen Å. Ellingsen's document "The stuff that makes a decent thesis", written for MSc students.

1 Overall considerations

Here are some considerations for how you should write and structure your report (for many if you this will be the first time writing a big report!). This is also what the examiners will think about when they judge how good the report itself is. It is recommended to use L^AT_EX; easy and fun to use is Overleaf (you might use the NTNU template, or any template you find appropriate for your report). If you find L^AT_EX too challenging you might use MS Word. For more tips on report/thesis writing, see SokOgSkriv.

Outline of present document: the present section describes points concerning the overall structure and content of the thesis; in § 2 a list of important elements concerning the report structure is listed; and § 3 includes considerations on the content of the different sections.

1.1 Implementation of the two models from subgroups

The report should be written by the whole group, but both models from subgroups **must** be implemented in the *Methodology* chapter. See § 3.6 for more details.

1.2 How long must the report be?

6 000 ±5% words, with the following rule.

Excluded in word counting:

- Cover or Title Page

- Abstract
- Contents Page
- List of Abbreviations / Acronyms
- List of Tables/Figures
- Tables
- Figures
- Reference List and/or Bibliography
- Appendices

Included in word counting:

- Main Text e.g. Introduction, Methodology, Results, Discussion and Conclusions.
- Chapters headings and subheadings
- Footnotes (relating to in-text footnote numbers)

Be aware that:

- Word count should be written in the front page.
- If the word count falls outside the specified range, the grade will be reduced by one level.

1.3 Delivery

Deliver your report on Canvas in .pdf. Name of the file should be "NN_Report.pdf", with NN= group nr.

1.4 Use of AI

You must write which and how AI assistance you used, as subchapter in Preface.

For more info see NTNU rules here: Artificial intelligence in student papers.

1.5 Grading scale

Use the present document to get rules and guidelines on report writing. The grading form used for grading the report is shown in Fig. 1 For a more general grading scale see NTNU's grading scale.

		Weighted	Points											
	Evaluation criteria	points	Weight	10	9	8	7	6	5	4	3	2	1	0
1	Structure of the report	0	1	0	0	0	0	0	0	0	0	0	0	0
2	Abstract and Introduction	0	1	0	0	0	0	0	0	0	0	0	0	0
3	Theory	0	2	0	0	0	0	0	0	0	0	0	0	0
4	Method	0	2	0	0	0	0	0	0	0	0	0	0	0
5	Results	0	2	0	0	0	0	0	0	0	0	0	0	0
6	Discussions	0	3	0	0	0	0	0	0	0	0	0	0	0
7	Conclusions	0	1	0	0	0	0	0	0	0	0	0	0	0
8	Problem solving / creativity / Engineering approach	0	1	0	0	0	0	0	0	0	0	0	0	0
9	Literature study / references	0	1	0	0	0	0	0	0	0	0	0	0	0
10	Utilisation of available resources / cooperative skills	0	1	0	0	0	0	0	0	0	0	0	0	0
11	Language	0	1	0	0	0	0	0	0	0	0	0	0	0
12	Overall evaluation	0	1	0	0	0	0	0	0	0	0	0	0	0
Sum weighted points		0	17											
Average weighted points / Grade		0,000	F											
Suggested grade		Examiner 1		A		10,00 - 8,850								
		Examiner 2		B		8,849 - 7,650								
		Final Grade		C		7,649 - 6,450								
				D		6,449 - 5,250								
				E		5,249 - 4,050								
Comments				F		4,049 - 0								

Check this out for examples SokOgSkriv citations.

References to accessible and original sources are preferable. If this is not possible, you might refer to internal documents (as the course assignments). In that case, check how to refer to internal documents for the citation style you chose.

1.10 Report everything you've done

The time available for your project was limited, so it's important that you report everything you have done. *Let's say that you tested 20 airfoils in order to find the one giving highest C_l/C_d , well...write it.* You definitely don't need to show all the polar curves, but you can write in your narrative that you tested 20 airfoils. And if you somehow want to show also the polars of the different tested airfoils, so the Appendices is the right place, but then remember to refer to them in the main text.

1.11 Coherence, structure and content of sections

Remember that a work must be reported properly. A good work, but badly reported gives bad grades. It's important for your project to be *coherent* from start to finish, that is, that all the different parts have a natural place as part of a whole, and language is consistent throughout the text. Tips on structure and content of each section of the report are given, respectively, in § 2, and 3.

It is a good thing to *point out explicitly* how the parts relate to each other, and in particular how each part relates to the stated objectives and task. In ch. 3, say, could say things like *"It will now be presented the methodology used to modify the blade, designed with BEM in section 1.3, in order to follow the restrictions on production. The comparison between the performances of the modified blade will be reported later in section 4.4"*. Or something like that.

Each of you will, most luckily, write just one part of the report. That is alright, but it is important that, once the report is finished, one (preferably more) of you read through the report before handling it in, to check for:

- Coherence
- Structure
- Content of each section

2 Structure of the report

List of important points on structure of the report (use it as a check list).

- Sections and chapters sequence and numbering: Preface, Summary (or Abstract), Table of contents, List of symbols, Terms list (optional), 1. Introduction, 2. Theory, 3. Materials and methods, 4. Results 5. Discussion (or 4. Results and Discussion), 5./6. Conclusion.
- Figure caption below the figure
- Table caption above the table
- Post-process figures (results obtained with Excel/Matlab/Python): White background (possibly guides), no frame around the whole figure, name on the axis, relevant number of significant digits, legible axes.
- All graphs, equations and tables must have a number.
- All graphs, equations and tables must be referenced, with numbers (eg equation 2.1), and explained/mentioned in the text.
- use hyperlink
- All symbols must be explained the **first time** they are introduced (even if the list of symbols is included).
- With a possible list of symbols, all symbols must be included and contain explanations with units. Also Greek and Latin letters must be separated (e.g. before the Greek and after the Latin), and be in alphabetical order.
- use units, use units, use units!
- Be consistent with symbols (*don't use V =wind speed, and later U =wind speed*).
- An appendix must be understandable without reading the report.
- An appendix must be marked with letters not numbers.
- The report must be understandable without reading appendices.
- Calculations must not be shown in the results. If desired, they must be presented in the appendix (*Ex. don't show $x=4/2=2$, but just $x=2$*).
- (if using L^AT_EX) Equations: use \ left(and \ right).
Ex. using those: $\left(\frac{x^2}{2}\right)$. Ex. NOT using those: $(\frac{x^2}{2})$

- All numbers and digits must be presented with units (did I already write that?).
- Always deliver as a PDF.
- Everything citable must be cited, as statements and figures taken somewhere else.
- Is missing stuff? (It should not)
- Is superfluous material avoided? (It should)
- Is it easy to see how different sections are related? (It should be)

3 Content of each section

Here are some more considerations on the different sections. Always remember that everything must serve a purpose in your overall argument. The contents of each section are often flexible in terms of how you write it and in what order. The guiding principle is always: try to write it so that it “tells your story” as clearly as possible.

An important rule is always that *the Abstract and Conclusions should be stand-alone*, that is, they can be read without having to read anything else first!

3.1 Preface

Checklist:

- Are the contributions of each of you to the project work included?
- Is the type of assignment and the scope of the assignment well enough described?
- Are supervisors, employees and others who have contributed to the execution of the work thanked?
- Are the names of the group members listed in the preface and signed (with date)?

3.2 Abstract

The abstract is a mini-summary of your thesis without the detailed arguments. It should say 1) what you did and how, 2) what you found and 3) what you concluded from it.

3.3 Symbol list (if included)

Checklist:

- Are all symbols from equations (and possibly other) included in the symbol list?
- Are all the symbols explained briefly, concisely and precisely, and the units included?
- Are the Greek and Latin letters separated (e.g. before the Greek and after the Latin), and in alphabetical order?

3.4 Introduction

The introduction has several purposes:

1. Motivate the work: why is it interesting and important?
2. Provide the background (e.g., *what is the development of wind energy?*)
3. State the objectives of the project.
4. Introduction should end with a brief outline of the rest of the thesis. The idea is once again to show how it all fits together.

3.5 Theory

The Theory chapter contains anything the reader needs to know to understand what you have done. You have to use some sound judgement here. It is a common mistake to get a bit carried away here and include material that's beside the point. **Another common mistake is to repeat derivations that can be found in a text book.** Unless it's your own derivation, just give the resulting equation and explain what it means, but refer to the textbook (all the written equations in the theory must indeed be referred to the source). *You don't need to write the whole actuator disk theory, since it is simple to refer to a book.*

Reference to textbooks/original sources are to be preferred, rather than compendium and presentations.

In addition, you should answer 'yes' to the following questions:

- Does the theory part cover the points raised in the discussion well enough?
- Does the theory chapter have a logical and appropriate structure?
- Are all the opinions/beliefs omitted?

3.6 Methodology

Your methodology must be explained carefully enough to allow others to reproduce what you have done, without needing to guess your reasoning or choices. This is an important scientific principle. Beyond presenting the technical aspects, the key task is to make the case that *your results can be trusted*, meaning that they are correct and support your conclusions.

In practice, this means describing your methods and reasoning in sufficient detail while avoiding unnecessary step-by-step explanations of standard software operations (for example, QBlade procedures that are already covered in the user manual). You should therefore:

- Describe both numerical and experimental procedures thoroughly and coherently (avoid using bulleted lists for this part).
- Focus on what you did and why, not on every step or button click.
- Provide key input parameters, limitations and assumptions specific to your case.
- Reference standard workflows when appropriate, e.g.:
“The aerodynamic performance was simulated in QBlade following the standard BEM workflow (see QBlade User Manual).”
- Use explanatory figures, tables, or summaries to present input and output data, but avoid step-by-step sequences unless they clarify a non-standard process.

In short: Describe your reasoning and specific choices in detail, but only outline standard procedures that can be found in manuals or textbooks.

3.6.1 Implementation of both groups’ models

The models of both subgroups should be presented, in order to show the work done by all the members.

If the method used by the two subgroups is common, you may describe it as a shared procedure.

Include only the numerical results from both models that contribute to your discussion. E.g., comparing the two total efficiencies justifies the choice of the model for production, but simply showing the $C_P - \lambda$ curves for both models does not provide much value unless you also interpret or discuss what the curves reveal. If you have meaningful insights or comparisons to highlight, then it is appropriate and useful to include the figures.

3.7 Results and Discussion

Typically you report your results first (in the Results section) in the form of graphs, tables, etc, and then discuss them after (in the Discussion section). One can also structure it so that some results are reported and then discussed, then moving to the next batch of results. Whichever makes the best text. However, it should be well structured and easy to follow (the reader should not have to flip back and forth through the report). However, the proper way to do it is to *first* present the results objectively and free of interpretation and *then* discuss the implications.

Checklist for the Results section/part:

- Do the results make sense?
- Are the results presented clearly?
- Is it explained where all the numerical values originate from (from which equations), based on Theory?
- Are the details of the calculations omitted (they should)? Ex., don't show $x=4/2=2$, but just $x=2$.
- Are all the opinions/beliefs omitted (they should)?

Checklist for the Discussion section/part:

- Are all the presented results discussed?
- Has the authors come up with their own argumentation and independent reflection?
- Are all topics introduced in the theory discussed?

3.7.1 Future outlook

At the end of the Discussion it is a good idea to include a short paragraph with few ideas for further work.

3.8 Conclusion

Be aware to restate your findings. Your Conclusion should NOT introduce new arguments and/or new data. Checklist:

- Is the conclusion written short and concise, with reasonable length and sober language?
- Are conclusions drawn from the discussion (nothing new should be written in the conclusion, in terms of data, ideas and citations)?

- Are the most relevant findings, assessments and conclusions presented?
- Does the conclusion have a reasonable length in relation to the content?
- Does the report bites its own tail (all the themes and issues that you raised in your introduction must be referred to again in one way or another)?

3.9 Bibliography

All entries in the bibliography must be explicitly cited in the main text. There are two standard formats in our fields: numbered by order of appearance, [1], [2], ..., or alphabetically by name and year “Townsend (1956)” etc.

Make sure to check that your bibliography is consistent and references are complete. It should be easy for the reader to find your references.

It is highly recommended to write your thesis in $\text{\LaTeX}$, which has the two styles as options and sorts out the bibliography automatically. You can find the references formatted for $\text{\LaTeX}$ directly in Google Scholar, but note that the formatting can differ, so edit them to get them consistent.

Should you for some unfathomable reason wish to write in MS Word, EndNote is a useful tool for keeping references in shape.

3.10 Appendices

Appendices are entirely optional. An appendix is a place to put material you think should be available, is relevant and interesting, but doesn’t have a natural place in your coherent argument. It could be extra results you got, perhaps further details on your numerical or experimental set-up, computer code or data. It could be lengthy calculations or long technical descriptions which would clog the main text inordinately. Make clear in the appendix which part of the main text it is linked to, and refer to the appendix in the main text, typically with “further details can be found in Appendix A” or some such.